AI.409

Robot Simulation

COURSE OVERVIEW

Virtual robotics: simulators, physics engines, digital twins, and sim-to-real transfer.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: VIRTUAL ROBOTS

KEY TOPICS

- Why simulate
- Simulators vs. real robots
- Digital twins
- Sim-to-real gap

TEACHING FOCUS

Test robots without risk.

KEY TAKEAWAY Simulation is the robot's practice ground.

ROBOT SIMULATION

2

SESSION

60 MINUTES

PHYSICS ENGINES

KEY TOPICS

- Rigid-body dynamics
- Contact and collision
- Actuator models
- Gazebo, PyBullet, MuJoCo

TEACHING FOCUS

Simulate physics faithfully.

▲ **KEY TAKEAWAY** Physics engines make virtual motion real.

ROBOT SIMULATION

3

SESSION

60 MINUTES

SIMULATION WORKFLOW

KEY TOPICS

- URDF/SDF models
- Worlds and environments
- Sensors in sim
- Logging and replay

TEACHING FOCUS

Build simulation scenes.

KEY TAKEAWAY Models and worlds define the sim.

4

SESSION

60 MINUTES

SIM-TO-REAL TRANSFER

KEY TOPICS

- Domain randomization
- System identification
- Transfer learning
- Hardware validation

TEACHING FOCUS

Move from sim to real reliably.

▲ KEY TAKEAWAY The sim–real gap must be bridged.

ROBOT SIMULATION

5

SESSION

60 MINUTES

SIMULATION FOR AI & ROBOTICS

KEY TOPICS

- Training RL in sim
- Testing perception
- Regression testing
- CI for robotics

TEACHING FOCUS

Develop and test in simulation.

KEY TAKEAWAY Simulation accelerates development.

6

SESSION

60 MINUTES

ROBOT SIMULATION & REVIEW

KEY TOPICS

- Simulation project
- Course recap and assessment
- Toward robot vision (AI.410)
- Sim showcase

TEACHING FOCUS

Validate a robot in simulation first.

KEY TAKEAWAY Fail in sim, not in the field.